COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	9/1/26	**DATE SUBMITTED:**	9/1/26
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ACTIVITY NO. 7.2 **Algorithm Complexity**

1. PROCEDURE OUTPUT

1. Finding an Element

```
None
 for (int i = 0; i < n; i++)
 {
    if (arr[i] == target)
    {
       return i;
    }
}
```

Answer:

Best Time Complexity:O(1)

Worst Case Time Complexity: O(n)

Explanation: Best case happens after finding the target on the first try, worst case on the other hand happens if the target was not found on the first try which executes the loop for n times.

2. Printing All Elements

```
None
 for (int i = 0; i < n; i++)
 {
    cout << arr[i] << endl;
}
```

Answer:

Best Time Complexity: O(n)

Worst Case Time Complexity: O(n)

Explanation: The loop runs at a number of n times.

3. Nested Loops

```
None
 for (int i = 0; i < n; i++)
 {
    for (int j = 0; j < n; j++)
    {
       cout << i << " " << j << endl;
    }
}
```

Answer:

Best Time Complexity: $O(n^2)$

Worst Case Time Complexity: $O(n^2)$

Explanation: After each turn of the outer loop, the inner loop runs n times (n x n) which totals in n^2 each step.

4. Conditional Processing

```
None
 if (n > 100)
 {
    for (int i = 0; i < n; i++)
    {
       cout << arr[i];
    }
 }
 else
 {
    cout << "Small input";
}
```

Answer:

Best Time Complexity: O(1)

Worst Case Time Complexity: O(n)

Explanation: Best case happened after the condition is satisfied on the first try, and worst case happens if it was not satisfied on the first try.

5. Fixed Inner Loop

```
None
 for (int i = 0; i < n; i++)
 {
    for (int j = 0; j < 10; j++)
    {
       cout << i << j;
    }
}
```

Answer:

Best Time Complexity: O(n)

Worst Case Time Complexity: O(n)

Explanation: The inner loop is running for ten times and for every step (10 x n), after the constants are dropped it is simplified to O(n) each step.

6. Two Consecutive Loops

```
None
 for (int i = 0; i < n; i++)
 {
    cout << arr[i];
 }

 for (int j = 0; j < n; j++)
 {
    cout << arr[j] * 2;
}
```

Answer:

Best Time Complexity: O(n)

Worst Case Time Complexity: O(n)

Explanation: The code runs two different loops for n times which prints the value of i. The steps are doing n + n total of steps each time it was executed.

7. Dividing by Two

```
None
 int x = n;

 while (x > 1)
 {
    x = x / 2;
}
```

Answer:

Best Time Complexity: O(log n)

Worst Case Time Complexity: O(log n)

Explanation: The x is divided in two each step which takes log n steps for every step.

8. Nested Loop With Different Limits

```
None
 for (int i = 0; i < n; i++)
 {
    for (int j = 0; j < i; j++)
    {
       cout << i << j;
    }
}
```

Answer:

Best Time Complexity: O(n²)

Worst Case Time Complexity: O(n²)

Explanation: The first loop runs for the n time and the j runs for the same time until the condition on their loops is satisfied.

9. Early Termination

```
None
 for (int i = 0; i < n; i++)
 {
    if (arr[i] == target)
    {
       cout << "Found";
       break;
    }
}
```

Answer:

Best Time Complexity: O(1)

Worst Case Time Complexity: O(n)

Explanation: Best case happens after finding the target on the first try which causes the program to break.

Worst case checks every single element to find the target.

10. Two-Level Search

```
None
 for (int i = 0; i < n; i++)
 {
    for (int j = 0; j < n; j++)
    {
       if (arr[i] == target)
       {
          break;
       }
    }
}
```

Answer:

Best Time Complexity: O(n)

Worst Case Time Complexity: O(n²)

Explanation: Best case happens if the condition was satisfied on the first try. Worst case happens if the program never finds the target (n x n = n²).

11. Linear Loop With Logarithmic Inner Work

```
None
 for (int i = 0; i < n; i++)
 {
    int x = n;

    while (x > 1)
    {
       x = x / 2;
    }
}
```

Answer:

Best Time Complexity: O(n log n)

Worst Case Time Complexity: O(n log n)

Explanation: The outer loop runs n times, and the while loop divides x by 2 until x <1.

12. Conditional Nested Loop

```
None
 if (arr[0] == target)
 {
    cout << "Found";
 }
 else
 {
    for (int i = 0; i < n; i++)
    {
       for (int j = 0; j < n; j++)
       {
          cout << arr[i] + arr[j];
       }
    }
}
```

Answer:

Best Time Complexity: O(1)

Worst Case Time Complexity: $O(n^2)$

Explanation: Best case happens if the condition was satisfied after the first try (arr[0] == target). Worst case happens if the first value does not satisfy the condition which runs the program's nested loop.

13. Finding the Maximum

```
None
 int max = arr[0];

 for (int i = 1; i < n; i++)
 {
    if (arr[i] > max)
    {
       max = arr[i];
    }
}
```

Answer:

Best Time Complexity: O(n)

Worst Case Time Complexity: O(n)

Explanation: The for loop will repeat and check every number therefore it will stay as O(n).

14. Three Nested Loops

```
None
 for (int i = 0; i < n; i++)
 {
    for (int j = 0; j < n; j++)
    {
       for (int k = 0; k < n; k++)
       {
          cout << i + j + k;
       }
    }
}
```

Answer:

Best Time Complexity: $O(n^3)$

Worst Case Time Complexity: $O(n^3)$

Explanation: The three nested loop runs n times, n x n x n = n^3.

15. Comprehensive Analysis

```
None
for (int i = 0; i < n; i++)
```

Answer:

Best Time Complexity: O(1)

Worst Case Time Complexity: O(n log n)

Explanation: Best Case complexity happens after satisfying the first step which is if (arr[i] == target) it will return i. Worst case happens if the given condition is not satisfied on the first try, this executes the while loop where it runs log n times.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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